

Aryan Miriyala

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PROFESSIONAL SUMMARY

Software engineer focused on applied AI, backend/full-stack systems, and production-minded reliability across **Python, TypeScript, C++, SQL, Node.js, Next.js, PostgreSQL, Docker, and AWS**. Built AI-assisted engineering workflows, secure internal tools, LLM evaluation pipelines, data automation, and systems projects with emphasis on maintainable architecture, clean code, privacy, and end-to-end ownership.

EDUCATION

Bowling Green State University

Master of Science in Computer Science, GPA: 4.00/4.00

Bowling Green, OH
Aug. 2024 – Aug. 2026

Bowling Green State University

Bachelor of Science in Computer Science, GPA: 4.00/4.00

Bowling Green, OH
Jan. 2021 – Apr. 2024

EXPERIENCE

Artificial Intelligence & Computer Science Intern

SmartSolve Industries

May 2026 – Present
Bowling Green, OH

- Applied **Codex** and **Claude Code** as AI coding agents for planning, implementation, debugging, and self-review while keeping proprietary-code isolation and human review central.
- Built secure internal software with **Next.js, TypeScript, PostgreSQL, Drizzle ORM, SSO**, and auth middleware for onboarding and org-chart workflows around sensitive employee data.
- Created a cross-platform **Docker** devcontainer for **WSL2/Colima**, reducing setup friction and standardizing reproducible environments for AI-assisted development.

Data Engineering Intern

American Association of Insurance Services

May 2025 – Aug. 2025
Lisle, IL

- Automated a fully manual **SQL** billing workflow with **Python, PySpark**, and **AWS Glue**, processing **20+ TB** of production insurance data for **700+** member companies.
- Built validation, reverse-ETL, and custom **AWS IAM** access patterns across **S3** and MDM workflows, improving controlled data access and reducing recurring manual operations.

Software Engineering Intern

American Association of Insurance Services

May 2023 – May 2024
Lisle, IL

- Engineered **Python**/JSON backend automation that generated **1,000+** production **SQL** tables across **Oracle/PostgreSQL**, supporting standardized data models across **10+** insurance lines.
- Built **Python/Pandas** workflows with multiprocessing, fuzzy matching, and hashing to clean large CSV datasets for production data-standardization pipelines.
- Contributed to a **React/Node.js** modernization by implementing **RBAC** and **JWT** patterns, replacing broad employee-wide access with role-specific controls for roughly **110** internal users.
- Automated database validation with **AWS Lambda** triggers that tokenized PII using **hashlib/boto3** and ingested structured JSON into **S3** for downstream workflows.

Software Engineering Intern

Alliance for Paired Kidney Donation

May 2022 – Aug. 2022
Toledo, OH

- Secured an **AWS-hosted Lucee/CFML** healthcare workflow platform with global **anti-CSRF** protection, IP-aware audit logging, and access-denied/404 routing across form-heavy internal workflows.
- Modernized legacy form workflows with **HTML5** inputs, JavaScript fallbacks, themed CSS, and **CFML** updates to improve data-entry reliability for clinical operations users.
- Debugged and triaged **4** reliability issue classes across import ordering, MFI upload paths, duplicate-antigen page crashes, and reusable modal/tab screens to stabilize internal operations workflows.

PROJECTS

Diff-Grounded PR Description Generation | *Python, OpenAI API, GitHub API, LLM Evaluation*: Built an LLM pipeline generating evidence-grounded PR descriptions from commits, diffs, linked issues, and metadata, with structured context construction and automated evaluation against code-change evidence.

DreamScape | *React Native, Expo, TypeScript, Gemini API, ElevenLabs API*: Built an AI mobile app with document summarization, generated quizzes, text-to-speech learning cues, cached audio playback, cue scheduling, and local-first session tracking.

Self-Adaptive Parallelism | *C++, OpenMP, Python, Docker, Linux*: Built a C++/OpenMP runtime system using telemetry, scheduling algorithms, UCB optimization, and benchmarks across Mandelbrot, heat diffusion, and reduction workloads.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, C++, C#, SQL, C, Lucee/CFML

AI & Agents: Codex, Claude Code, OpenAI API, Gemini API, ElevenLabs API, LLM evaluation, RAG, prompt engineering
Backend, APIs & Auth: Node.js, Express, REST APIs, Next.js, React, PostgreSQL, Oracle, MongoDB, Drizzle ORM, JWT, RBAC, SSO

Cloud, Systems & Reliability: Docker, devcontainers, Git, Linux, AWS Glue, Lambda, S3, IAM, OpenMP, runtime instrumentation, audit logging, anti-CSRF